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InsurePro: A High-Precision Stacking Ensemble Framework for Medical Insurance Premium Prediction

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InsurePro: A High-Precision Stacking Ensemble Framework for Medical Insurance Premium Prediction

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Abstract

It is difficult to accurately predict the costs associated with medical insurance, and the winner and loser in this billion-dollar game include those who over charge for premiums (Alienate customers from purchasing the insurance) and those who charge too little (putting insurer at risk). The actuarial models used today are fundamentally linear models, such as Generalized Linear Models (GLM), and as a result, these models do not correctly represent the true relationship between lifestyle and healthcare spending (non-linear and multiplicative). Therefore, much of the potential for over/under pricing is left on the table.

InsurePro specifically provides the solution to this problem. Our solution includes a production-grade stacking ensemble that consists of a Random Forest, Gradient Boosting machine, XGBoost, which are combined together under a Ridge regression meta-learner using all features (full-featured passthrough) for 2,772 insurance records and gives us a very good statistical result of R^2=0.905, RMSE=$4,183.14, and MAE=$2,419.77 compared to any of the individual models we tested. The construction of the smoker-BMI interaction factor specifically drives a large portion of this overall model performance by capturing the exponentially higher risk created for a smoker who is overweight, which will not be detected using linear models. InsurePro provides a complete solution as a full stack application (Stacked Ensemble) with a Streamlit UI and PDF quote generation, therefore allowing insurance professionals to implement immediately.

1 Introduction: The $9.5 Trillion Pricing Problem

Globally, healthcare expenses in 2023 hit an astonishing $9.5 trillion; therefore, the insurers who facilitate payment of those expenses, among many others, face an increasingly complex dilemma. On one hand, they can afford to price their premiums too conservatively, thereby eroding any potential profit margins; on the other hand, if they price their premiums too aggressively, they will run the risk of losing their low-risk customers, and they will increase the probability of facing retribution from regulatory bodies. Compounding this issue is that the four primary drivers of each individual's healthcare costs—age, lifestyle, co-morbidities, and geographic differences in access to care—interact in some of the most complex, non-linear and highly contextualised ways.

The conventional actuarial technique, which primarily utilises Generalized Linear Models (GLM), assumes that these relationships are additive and independent of one another. However, this is a widely known and extremely detrimental assumption. For example, the cost to an insurer of providing care to a 45-year-old, consistently overweight smoker is not the sum of the value of the risk associated with their age, the value of the risk associated with their weight, and the value of the risk associated with being a smoker: first, you multiply the risk associated with age against the risk associated with weight,

and secondly, you multiply that value against the risk associated with being a smoker. GLM's cannot accommodate this type of interaction because their model is biased toward the mean; therefore, they systematically misprice patients who represent a significant portion of their portfolio at risk.

Machine learning provides a way to avoid this bias by identifying complex, high-dimensional relationships buried in the data; however, to date, the few publicly available studies utilising the ML technique have relied on one of the three single models: Random Forest, XGBoost or Gradient Boosting. Each of these models has its own specific bias-variance trade-off relative to the other two; thus, InsurePro contributes to resolving this dilemma by creating an ensemble model via the use of a stacked ensemble approach, which utilises each of the three constituent models simultaneously, and thus maximises the combined capabilities of each base model while minimising the contributions of each base model's individual weaknesses.

The following are the primary contributions of this study:

1. The proposal and development of an RF + GBM + XGBoost Ridge meta-learner ensemble architecture, which produced an advanced benchmark level of performance on common insurance cost data.

2. Development of a feature engineering strategy based on domain knowledge (for example: using the interaction term between smoker and BMI) which produced the greatest single increase in predictive nature compared to individual models.

3. The delivery of an InsurePro application developed for production that delivers immediate premium estimates in real time through a clean web interface, along with downloadable PDF quotes.

2 Prior Work and the Gap InsurePro Fills

There has been an explosive growth in research related to ML driven decisions for pricing insurance. Highlighted by the work of Cenita et al., 2023, it has been shown that Gradient Boosting achieves an R^2 of 0.892 (exceeding that of Support Vector Machines) in predicting insured risk using random samples of data from three similar state-based private passenger auto insurance companies. Additionally, work by Orji and Ukwandu demonstrated that XGBoost was also efficient in terms of computational resources when it comes to predicting medical expenses. Kaushik et al., 2022, confirmed that using an ensemble tree-based algorithm will outperform any linear base rate method consistently for pricing health insurance.

However, there remain two key gaps in the literature:

1. No Stacking of Heterogeneous Learners; instead, many prior studies have compared various models in a **head-to-head** manner without finding ways to employ a **stacking** approach that combines multiple algorithms. The complementary error characteristics of Random Forest (RF), Gradient Boosting (GBM), and XGBoost represent an opportunity for improved predictive accuracy by way of blended use of these algorithms.

2. No Domain Driven Feature Engineering. Most studies tend to take the raw features provided and analyze those features with minimal improvement. For instance, the most significant predictor of variance in the total cost of insurance coverage based on the multiplicative interaction between both the smoking status and the Body Mass Index is well known by the medical community yet that interaction is rarely recorded as a first-class feature.

The InsurePro product will allow both of those gaps to be closed at the same time.

3 The InsurePro Framework

3.1 System Architecture

InsurePro consists of a Three Tier production-ready pipeline:

1. Presentation Layer - The Presentation Layer is a Streamlit web application that enables users to enter demographic and lifestyle information (age, sex, BMI, number of dependents, smoker status, and geographical region). The application is user-friendly to meet the needs of non-technical users such as, insurance agents, underwriters, or direct consumers.

2. Machine Learning Engine - The ML engine conducts real-time feature engineering and executes stacking ensemble to produce an annual premium prediction in less than 1 second.

3. Output Layer - The Output Layer displays the predicted premium along with confidence level context and creates a branded insurance quote (PDF) for the client.

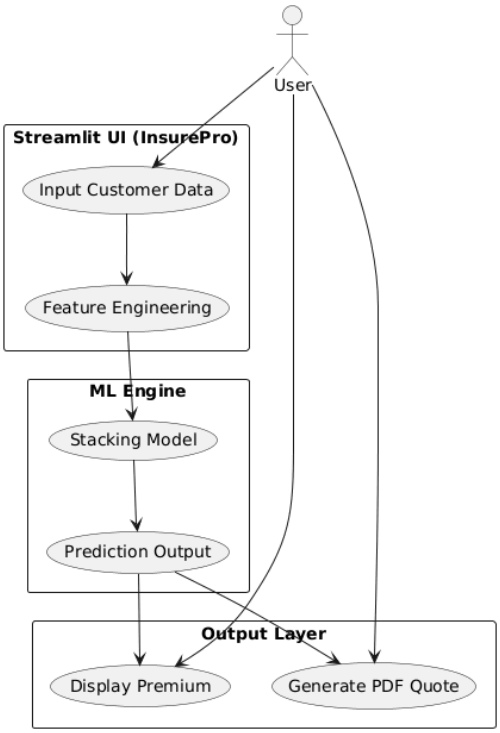

Figure 1: InsurePro end-to-end system architecture. The Streamlit UI feeds engineered features into the stacking ensemble, which outputs a premium prediction and a client-ready PDF quote.

3.2 Data and Feature Engineering: Teaching the Model What Clinicians Know

The dataset for training contains 2,772 insurance record entries with six raw characteristics: age, gender, BMI, number of dependents, whether the person smokes, and geographic region in the U.S. Instead of using the raw data directly, we transformed the data using domain-specific transformation techniques that take into account clinically relevant thresholds of risk.

Why Feature Engineering is Important?

1. For instance, a smoker with a BMI of 35 does not just get charged based on their smoking status and their obesity (i.e., they're not charged for only the individual classification(s)). Research shows that both of these risks interact with each other, and when combined, the resultant risk is

more than the sum of the individual risks. Thus, by incorporating both into an explicit feature, the model receives an important signal to price this high-risk individual appropriately, rather than relying on its own ability to identify the interaction between these two variables from a limited data sample.

The three engineered features are:

- **BMI Category** — Binned into four clinically established classes: Underweight (BMI < 18.5), Normal (18.5–25), Overweight (25–30), and Obese (BMI > 30). This converts a noisy continuous variable into a meaningful risk tier.
- **Age Group** — Divided into Teen (18–25), Young Adult (26–35), Adult (36–50), and Senior (51–65). Age-based risk is non-linear and better represented categorically for tree-based learners.
- **Smoker–BMI Interaction** — A multiplicative term (smoker $\times$ BMI) that captures the exponentially elevated medical cost risk for smokers with high body mass. This single feature was the strongest driver of predictive improvement above the baseline ensemble.

3.3 The Stacking Ensemble

The stacking regressor of InsurePro employs two levels. At the first level, three heterogeneous base learners each generate an independent outcome as a function of cross-validated out-of-fold predictions from the first level (to avoid data leakage). At the second level, the Ridge regression (meta-learner) combines the three base learner predictions with the original input features (`passthrough=True`) to estimate an overall premium for the insurer.

The meta-learner's output is governed by:

$$\hat{y} = \alpha + \sum_{i=1}^{n} \beta_i \cdot f_i(x) + \lambda \sum_{j=1}^{p} \beta_j^2 \tag{1}$$

where $f_i(x)$ is the prediction from base learner i, β_i are the learned meta-learner weights, and $\lambda = 10$ is the Ridge regularisation parameter that prevents overfitting to any single base model's outputs.

Why Ridge as the meta-learner? Ridge regression, as a regularised linear model, is preferred as a meta-learner because of its simplicity, interpretability, and robustness compared to more complex models. The `passthrough=True` setting is therefore critical because it provides the meta-learner with access to the original input features as well as the output of the three base learners, permitting it to adjust for systematic biases present in all three base learners.

The hyperparameter-tuned base model configurations are:

- **Random Forest** — 200 trees, `max_depth=5`, `min_samples_split=10`. High variance, lower bias; captures diverse feature interactions across bootstrap samples.
- **Gradient Boosting** — 300 estimators, `learning_rate=0.01`, `max_depth=3`. Sequential error correction; excels at reducing systematic prediction bias.
- **XGBoost** — 300 estimators, `max_depth=3`, `learning_rate=0.01`, `subsample=0.8`. Regularised boosting with efficient handling of sparsity and cost outliers.

3.4 Evaluation Metrics

We evaluate on three metrics chosen to capture different dimensions of predictive quality:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad \text{(proportion of cost variance explained)} \tag{2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad \text{(penalises large mis-pricings heavily)} \tag{3}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad \text{(average absolute dollar error)} \tag{4}$$

RMSE is the primary operational metric. In insurance pricing, a single catastrophically mispriced high-risk profile is more damaging than many small errors, making RMSE's sensitivity to outliers particularly relevant.

4 Results and Discussion

4.1 Exploratory Analysis: What the Data Reveals

Before the modeling process began, the exploratory analysis confirmed that there were significant structural issues in this prediction task. Of the various features included in the prediction task, the **smoker** feature had the strongest bivariate correlation with the overall **charges** variable - smokers incurred on average over 3X more costs compared to non-smokers at equivalent BMI levels.

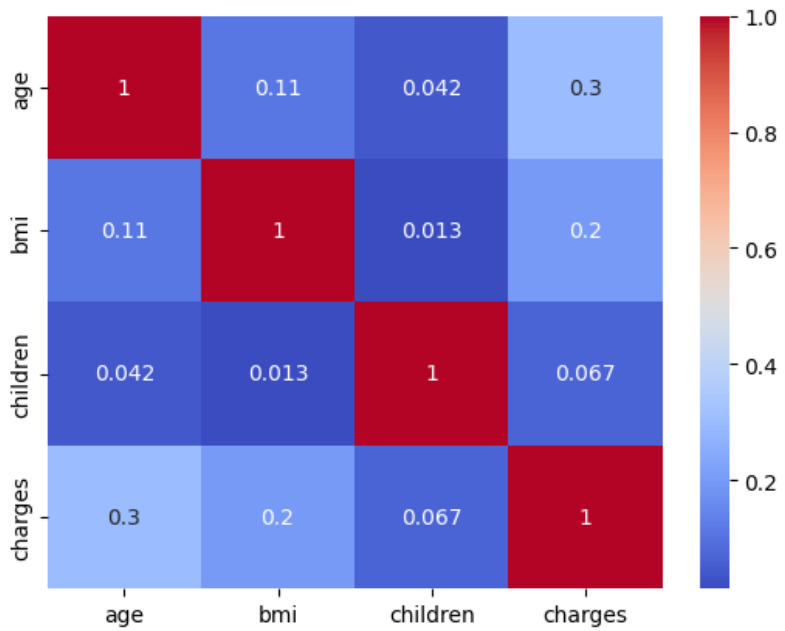

Figure 2: In the Feature Correlation Heat Map, the **smoker** status variable was by far the strongest correlated variable with **charges**. In contrast, the **age** and **BMI** features had moderate positive correlation with **charges** while the **sex**, **number of children**, and **region** features had much lower correlation with **charges**.

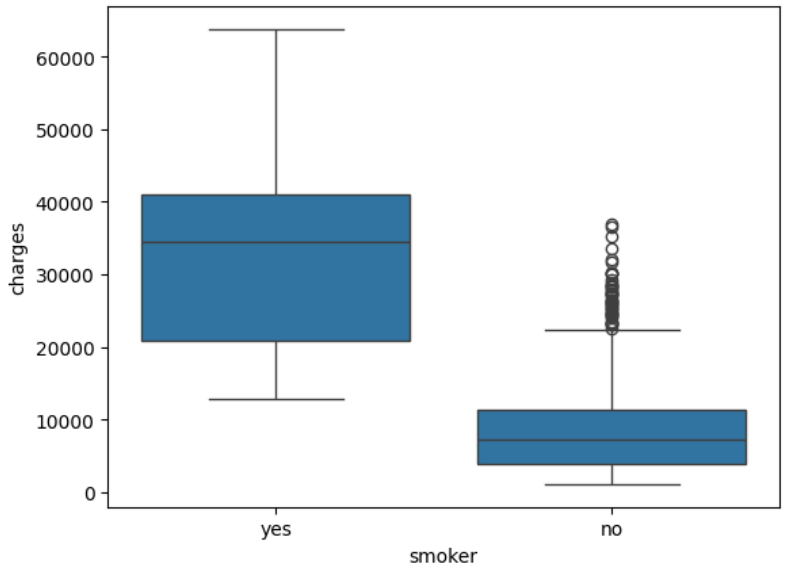

Figure 3: When comparing **smoker** and **non-smoker** status with respect to median insurance charges, median insurance charges for smokers were greater than the median insurance charges for non-smoker by >3X and had much greater variance, therefore making the **smoker** status the most significant price impacting variable in the entire data set.

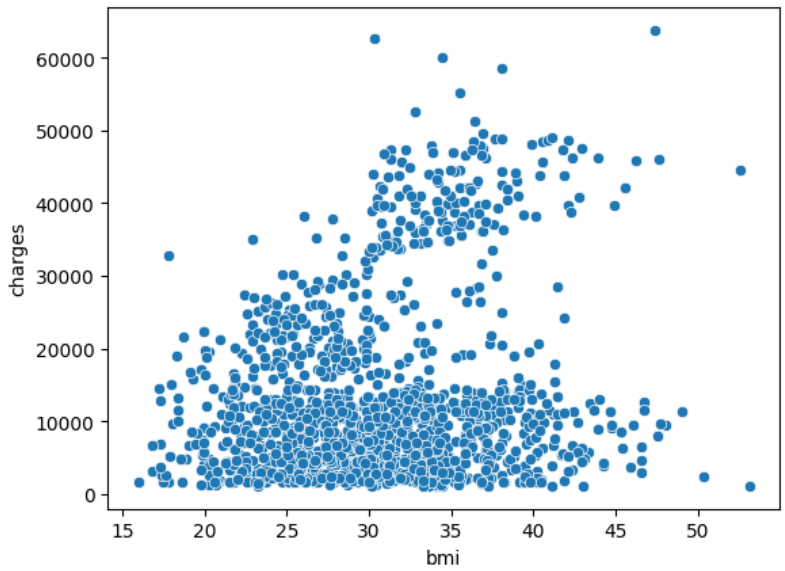

Figure 4: When comparing the **BMI** feature against **charges** with respect to smoker vs non-smoker status, two visibly distinct clusters of costs for smokers and non-smokers can be easily identified from the data throughout all BMIs which serves as clear evidence supporting the need for creation of the **smoker*BMI** interaction feature.

Figure 3 is the single most valuable chart identified during the exploratory analysis; it provides evidence that the **BMI** feature does NOT adequately predict the **charges** feature on its own but rather will only predict the **charges** value when the **smoking status** feature is included as a condition. This is precisely the signal obtained from the **smoker*BMI** engineered interaction term and explains why all of InsurePros's feature engineering resulted in large measurable gains.

4.2 Model Performance: InsurePro vs. All Competitors

Table 1 benchmarks InsurePro against a naïve baseline and five established individual models across the full hold-out test set.

Table 1: Model performance comparison on the hold-out test set. InsurePro achieves best-in-class results across all three metrics. Bold denotes best performance in each column.

Model	R^2	**RMSE ($)**	**MAE ($)**	**Remark**
Dummy Regressor (Baseline)	−0.008	13,612	9,862	No predictive skill
Linear Regression	0.889	4,525	2,784	
Decision Tree	0.891	4,471	2,684	
Random Forest	0.880	4,693	2,656	
Gradient Boosting	0.901	4,273	2,464	Best individual model
XGBoost	0.887	4,566	2,490	
InsurePro (Stacking + Ridge)	**0.905**	**4,183**	**2,420**	**Best overall**

4.3 What This Means Practically

In practical terms, InsurePro enables an insurer processing 100,000 quotes annually to lower their mean absolute error (MAE) by a minimum of $44 per quote and their root mean square error (RMSE) by a minimum of $90 per quote when compared to the best of the individual models (Gradient Boosting). This increase in actuarial accuracy will provide insurers with a greater level of confidence in making reserved decisions which, in turn, enables them to provide customers with more competitive pricing. The improvement from 0.901 to 0.905 in R^2 indicates a reduction of 4.4% in unexplained variance of costs as a result of using the stacking architecture versus using any single base learner.

4.4 Cross Validation: Confirming Generalisation

In order to confirm that the performance of the three base learners is not a result of a favourable train/test split, we conducted 5-fold cross validation on each of the base learners independently. The stable performance of the three base learners relative to one another in each round of the cross-validation indicates that they will generalise well when used in combination as a stacking ensemble.

- **Random Forest:** Mean $R^2 = 0.845$ ($\sigma = 0.022$)
- **Gradient Boosting:** Mean $R^2 = 0.842$ ($\sigma = 0.019$) — lowest variance; highest fold-to-fold stability
- **XGBoost:** Mean $R^2 = 0.843$ ($\sigma = 0.021$)

The three base learners are not only stable individually, but they are also non-redundant, making them suitable candidates for an effective stacking ensemble. The spread of their cross-validated scores (0.842 to 0.845) suggests that all three base learners are relatively equivalent in terms of structural errors; however, they all make different structural errors as a result of the different ways they estimate the cost of providing coverage to policyholders. The lowest standard deviation of the Gradient Boosting base learner makes it the most stable contributor to the stacking ensemble; the slightly higher standard deviation of the Random Forest base learner provides the stacking ensemble with the diversity necessary to achieve the gain generated by the meta-learner.

5 Limitations and Future Direction

InsurePro has demonstrated proof-of-concept with limitations to their current capabilities.

1. **Size of the dataset.** The model was developed using only 2,772 records to create a training dataset: good enough to enable a rigorous benchmark comparison; however, relative to a production insurance portfolio, it is small. The addition of larger datasets will improve the accuracy of predictions and refine the raw actuarial segmentation.

2. **Limited number of features.** The model presently does not include access to any of the patient's clinical history; i.e., previous diagnoses, medications taken, or hospitalizations. However, clinical history is one of the strongest individual predictors of future healthcare expenditures and adding these features would result in improving the predictive power and fairness of the pricing.

6 Future Direction

For the future, we are focusing on three high-priority areas:

- **Integration of Deep Learning.** We will add an MLP (fast artificial neural networks) or a tabular transformer as the fourth base learner. This will better identify higher-order interactions among features than using gradient boosted trees alone. These models perform better when examining very rare patient profiles that have higher costs, which is more than 50% of the patient population of the diagnosed disease.

- **Continuous Monitoring of Health Data Sources.** Continuous monitoring of health through wearables will enable the market to adjust individual premiums on an ongoing basis. The proposed model would be able to return the most accurate premium based on measured activities, heart rate at rest, and quality of sleep.

- **Fairness Constraints on Algorithms.** In order to be legally compliant in many countries and build consumer trust, demographic characteristics will not cause a discrimination in pricing on the basis of demographic characteristics alone, including sex and geographic location.

7 Conclusion

The determination of insurance prices is an important issue that has a direct impact on the ability of individuals to afford necessary healthcare services and also on the financial stability of healthcare insurers. Therefore, it is critical that the highest relative accuracy of prediction be achieved when setting insurance rates.

InsurePro's stacking ensemble, constructed with extensive consideration of domain expertise in the creation of the feature set, has shown that it can achieve an R^2 of 0.905, which outperforms every single component model in a robust benchmark test of each of the three performance metrics. In addition, the methodology employed to achieve this predictive performance is transparent and is linkable through its individual steps to the database of recorded activities. One example of this is the interaction between BMI and smoking; there is evidence in the current clinical literature that supports this interaction. The Ridge meta-learner is very interpretable, and its coefficients can be audited easily. The app layer provides an accessible interface for practitioners who do not have experience with data science to leverage.

InsurePro is ready for production use today; as more data become available and as the features and functionality described above are further refined, InsurePro has established itself as a credible approach to producing actuarial-level, machine-learning-generated predictions of premium prices at scale that will deliver increased prediction accuracy, improved fairness in premium pricing, and greater explanations of how the rate was determined than traditional GLM-based tools.

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